IDE 202.B

DESIGN MILESTONES

Merve Aral

2025-2026 Spring Semester

DESIGN THINKING PROCESS

Personal Fruit Tea Maker · Design Milestones Timeline

EMPATHIZE

Understanding the user

1

KEY DECISIONS

- Interviewed mother about abandoned Sage tea maker
- Posted on Reddit, collected real user responses
- Mapped universal pain points
- Researched existing products — gap confirmed

"I wouldn't buy a product just to boil some fruits."

DEFINE

Framing the problem

2

KEY DECISIONS

- Narrowed to employed solo dwellers
- Identified tea bag as real competitor
- Confirmed health + viral aspiration motivation
- Wrote design direction statement

"The real competitor is the tea bag."

IDEATE

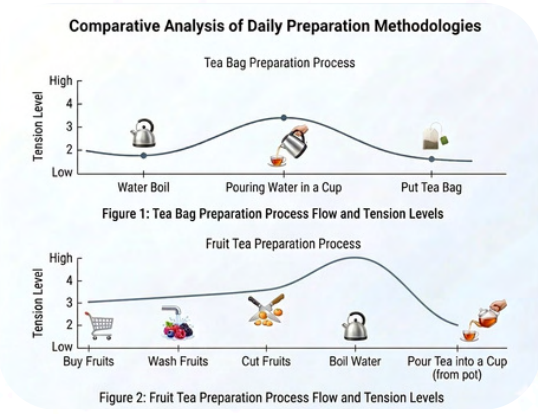
Generating concepts

3

KEY DECISIONS

- Drew before/after routine sketch
- Explored teapot form — rejected
- Explored kettle variant — rejected
- Selected single-vessel concept

"Boiling and drinking in one vessel."



PROTOTYPE

Making it tangible

4

KEY DECISIONS

- Built paper mockup to test proportions
- Decided on beer mug form
- Resolved filter, handle, lid system
- Generated renders in Vizcom & Gemini

"Wider than tall. Homey. Transparent."

TEST

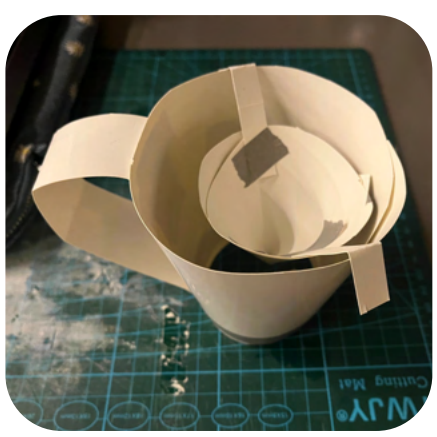
Getting feedback

5

KEY DECISIONS

- Teacher feedback: carry cup direction
- Teacher feedback: bigger glass, softer form
- Render review: steel too cold → warm tones
- Material and dimension refinement

"Concentrate on the fruit tea smell feeling."



KEYWORDS

Coreconcepts that shaped the design thinking

Research & User

Complexity aversion · Single serving · Employed solo dweller · Cognitive load · Planning barrier · Knowledge gap · Viral aspiration · Microplastics · Health motivation

Product Direction

Two-step experience · Ritual · Carry cup · Homey · Path of least resistance · Organic ingredients · Fresh fruit · Habit building

Form & Material

Double wall · Beer mug form · Transparent · Matte warm finish · Stainless steel · Rounded belly · Hidden compartment · Silicone seal · Borosilicate glass · Filter basket

CORE INSIGHT

> ***"The real competitor is not another tea maker. It is the tea bag."***
>
> The product must make fresh fruit tea the path of least resistance — removing the planning, transferring, and extra cleaning that currently makes a tea bag the easier choice. It must also feel homey and aspirational enough to become a habit, not just a purchase.

DESIGN DECISIONS

Key choices and reasonings throughout the process

1

PRODUCT TYPE

Single-vesselself-heating fruit tea maker

Collapsing boil, brew and drink into one vessel removes the biggest friction point.

2

INGREDIENTS

Freshfruit—not dried or freeze-dried

Health and nutritional value requires fresh. Dried fruit undermines the core purpose.

3

METAL MATERIAL

Stainlesssteelwith matte warm coating

Steel for heat conduction. Warm matte finish shifts feeling from industrial to homey.

4

LID

Starbucks-style slide opening

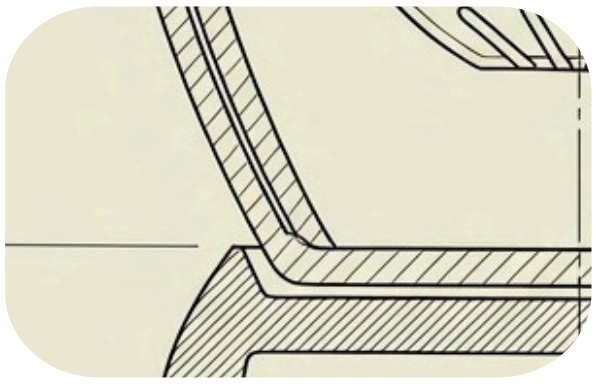
Familiar, one-handed, zero learning curve — critical for a time-poor user.

5

GLASS MATERIAL

Borosilicatedouble wall glass — middle section

Transparent, heat-safe, premium feel. Matches user trust and aspiration.



6

HANDLE

Rectangular compact — houses stick inside

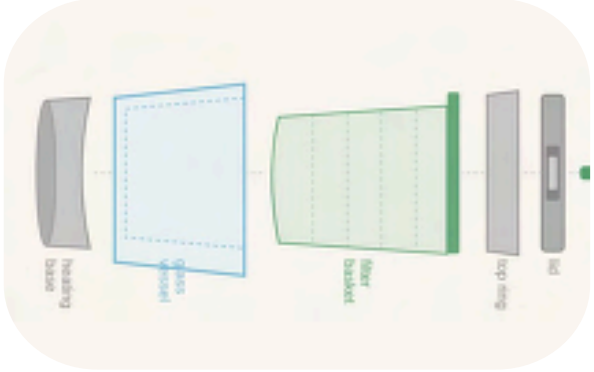
Minimal visually, functional grip, hidden compartment without extra components.

7

FILTER

Removable mesh basket on inner ledge

User can drink through it, remove or leave it in. Flexible without extra components.



SELECTED IMPACTFUL PROMPTS

Prompts that shifted or defined the design direction

RE-FRAMER

"I will design for people who crave fresh tea experiences by focusing on minimizing prep steps because complexity makes them default to tea bags."

AI RESPONSE

The statement was leaning toward a solution rather than a problem. Reframe toward the user's barrier: lack of knowledge, confidence, and time.

DECISION

Rewrote direction to focus on the user's gap rather than the product's action.

CRITIQUE

"I think if an employed person living alone wants to start a change like this, a product like this would encourage them."

AI RESPONSE

The product alone cannot motivate a habit. More interviews were needed to confirm that distrust of tea bags already exists in this user group.

DECISION

Sought Reddit comments to validate the assumption before building on it.

RE-FRAMER

"I couldn't find a specific step that people give up at."

AI RESPONSE

Abandonment happens before starting, not mid-process. The tea bag wins at the decision point. The real competitor is the tea bag.

DECISION

Shifted focus from process optimization to reducing commitment before starting.

EXPANDER

"I feel like it's impossible to make a product that makes the whole experience feel like two steps."

AI RESPONSE

"Two steps" is not literal — it describes how low-friction the experience needs to feel. Start with the single biggest barrier.

DECISION

Identified planning as the biggest barrier and began sketching.

VALIDATOR

"I draw a teacup with a boiling system — they won't need to waste time transporting the liquid."

AI RESPONSE

Confirmed: collapsing boil, brew and drink into one vessel removes real friction. Must feel distinct from a kettle the user already owns.

DECISION

Committed to the single-vessel concept as the selected direction.

EXPANDER

"I liked the render but I didn't like the steel look — it doesn't look authentic."

AI RESPONSE

Brushed steel feels industrial and cold, conflicting with the warmth of fresh fruit. Suggested matte warm coating rather than changing the material itself.

DECISION

Shifted toward homey finish — matte warm tones over stainless steel.

AI ROLE TAGS

How Ai functioned across different stages

Clarifier

Made vague feedback precise and actionable

- Broke down teacher feedback on form, handle and sensory direction
- Clarified borosilicate glass vs double wall insulation
- Distinguished thick vs big for the handle design
- Explained the material junction between glass and steel

Expander

Broadened thinking with new angles

- Introduced viral aspiration alongside health motivation
- Suggested carry cup direction fits the employed user lifestyle
- Connected smell/ritual feedback to organic form language
- Explored freeze-dried fruit as a middle ground

Validator

Confirmed decisions were sound and well-supported

- Confirmed single vessel concept collapses real friction
- Validated borosilicate glass for safety and premium feel
- Confirmed silicone seal at junction is industry standard
- Validated carry cup direction against user research

Re-framer

Challenged assumptions, offered new perspective

- Reframed direction from solution-focused to problem-focused
- Identified the tea bag as the real competitor
- Shifted focus to the pre-start decision moment
- Reframed fresh fruit storage as a behavior problem

Critique

Challenged ideas and pushed back on assumptions

- Challenged behavior change assumption before it became a premise
- Flagged aspiration drives purchase but not habit
- Identified fresh fruit storage as unresolved tension
- Pointed out kettle alternative wouldn't justify purchase

MY REFLECTIONS

Personal reflections on the process

RESEARCH

Throughout my research, I first tried to find problems in the wrong places, fishing out answers from the wrong people on social media. It made me waste time, but also helped me narrow down my user group. This way, the wrong answers lead me to the correct one. I defined the problem and the user group without thinking of the product or any solutions.

IDEATE

I was new to this new way of design process, so I had my struggles while trying to think ahead. Usually, I think in a more final product-focused way while designing. That's why when it came to ideate, I couldn’t sketch at first, stared at the blank page, trying to come up with a form. I talked with Ai as a partner. It suggested that I draw scenarios instead of the product, saying it will help me shape the form. I was surprised when it worked. I found myself drawing the product in that scene. Throughout the entire project, I can confidently say that AI provided the most significant assistance at this moment.

PROTOTYPE

I did a physical mockup first for AI to understand my product better, because with just sketches, AI can't render accurately. However, after that, I understood it was a struggle to make AI a product render despite doing rough sketches and mockups. I learned I shouldn’t trust AI saying, “It will fix it.” ” It will add it for me,” etc., because it doesn't. No matter how detailed you explained in your prompt, it's difficult to make AI understand something visually.

TESTING

AI was a good assistant at this stage. It helped me to analyse feedback and criticism, to help me make design decisions. I couldn’t test the final product physically. That's why AI helped me to make the best decisions for real-life scenarios. If I had more time, I would do a more detailed physical mockup and test it myself, but at this stage, I had to trust AI.

WHAT IF...

If I could do this project again, I would spend more time on my research. I would start with 3D modeling instead of letting AI do it in the first stages. This way, I could have more time to edit my model. I wouldn't use AI for generating images till the last stage, where my model would be ready to render, but as a thinking partner, I would use it in the whole process.

ACCEPTED · REJECTED · TRANSFORMED

ACCEPTED ✓

Carry cup lid direction
Teacher feedback + user research.
Employed user can take it to work.

Hidden stick in handle
Solves filter interaction without extra components.

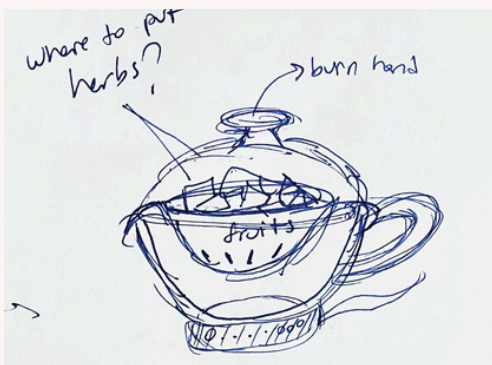
Single vessel concept
Boil,brewand drinkinone vessel. Core product logic confirmed.

Beer mug proportions
Wider than tall,belly form.Homey and generous.

Borosilicate glass
Heat safe, transparent, premium. Matches user trust and aspiration.

REJECTED ✗

Teapot form
Reintroduces a pouring step. Contradicts the core goal.



Fruit-compartment kettle
User already owns a kettle.Too similar to justify purchase.

Freeze-dried fruit
Under mines the fresh and nutritious purpose.

Beeswax wrap solution
Adds a step rather than removing one.

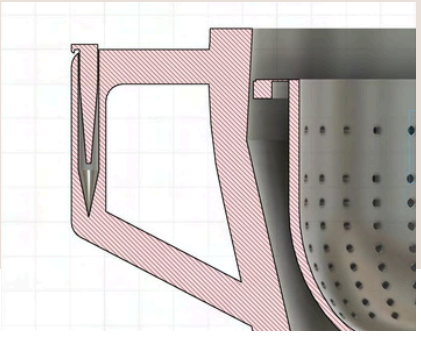
Full stainless steel finish
Feels industrial and cold.Conflicts with homey direction.

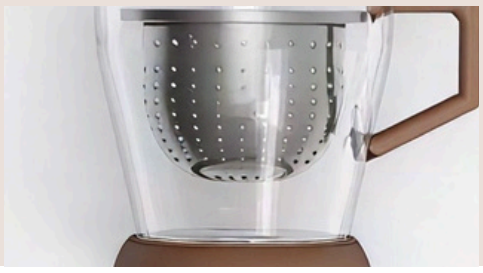
TRANSFORMED ↻

Large loop→**Rectangular compact**
Compact block with internal depth for stick.

Flat lid→**Slide opening**
Starbucks-style — familiar and one-handed.

Small glass→**Dominant glass**
Fruit and colour become the main visual story.





FINAL DIRECTION

Where the design stands

Design for employed people living alone who want to move away from commercial tea bags for health reasons but lack the knowledge, confidence, or time to make the switch. The product must make fresh fruit tea the path of least resistance — removing the planning, transferring, and extra cleaning that currently makes a tea bag the easier choice. It must also feel homey and aspirational enough to become a habit, not just a purchase.

Product Specifications

Form	Beer mug proportion, rounded belly, wider than tall
Total height	10–11 cm
Glass zone height	8 cm — dominant visual zone
Top opening (outer)	12–13 cm
Volume	550–600 ml total
Glass material	Borosilicate double wall · 3.5mm outer + 7mm air gap + 3.5mm inner
Metal finish	Matte warm coating on stainless parts
Handle	Rectangular · 4cm tall · 2.5cm wide · 3.5cm deep · internal stick slot
Lid	Slide opening · stainless + silicone seal
Filter	Removable mesh basket · rests on inner ledge 2cm from top
Heating dock	15cm diameter · 2cm tall · LED indicator

Open Questions/Opportunities

Fresh fruit sizing — how to guide users on preparation without adding cognitive load

3D modelling and physical prototype for ergonomic testing

User testing the handle grip and lid interaction with real users

Confirming 550–600ml volume feels satisfying after fruit displacement